

WORK PROCEDURE

PRO-MEC-006A

PROCEDURE FOR CHANGING A SAVER SUB

GOAL

The purpose of this procedure is to provide safe methods for effecting the change of the Saver Sub (Gear Box ITH)

APPLICATION

This procedure applies to all employees of MACHINES ROGER INTERNATIONAL.

INTERPRETATION

The staff with authority for the interpretation of this document are the Maintenance Manager and the Operations Manager. Any request for modification, revision or update must be made to one of these persons.

RESPONSIBILITIES

The worker is responsible for applying one of the following two methods when changing the Saver sub.

PROCEDURE

METHOD #1

1. Orient the boom horizontally.
2. Install the wrench on the sub saver.
3. To ensure that the wrench does not fall to the ground or is projected, a drill pipe is used and is installed as follows:
 - The female end of the rod is fixed in the jaws of the table.
 - Insert the other end of the drill rod into the gearbox saver sub by moving it without rotation. DO NOT SCREW THE ROD TO THE SAVER SUB
4. Once the drill rod is firmly seated in the saver sub, turn the rotation counter-clockwise until the sub saver is unscrewed. Use the hydraulic force to completely unscrew the saver sub. It is possible that it will require hitting wrench with sledgehammer

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5. Move the gear box a little, remove the saver sub
6. Remove the teflon protector from the saver sub to reinstall on the new saver sub.
7. Screw the new saver sub by hand for a few turns only.
8. Install the wrench on the sub saver and reinstall the drill pipe as described in step 3.
9. Once the drill rod is firmly seated in the saver sub, turn the rotation clockwise. Use hydraulic force, previously adjusted to 2800 psi, to fully screw the saver sub to the gear box contact.

METHOD #2

1. Position the boom vertically.
2. Place the key on two "jaw" centralizers previously placed on the boom tables.

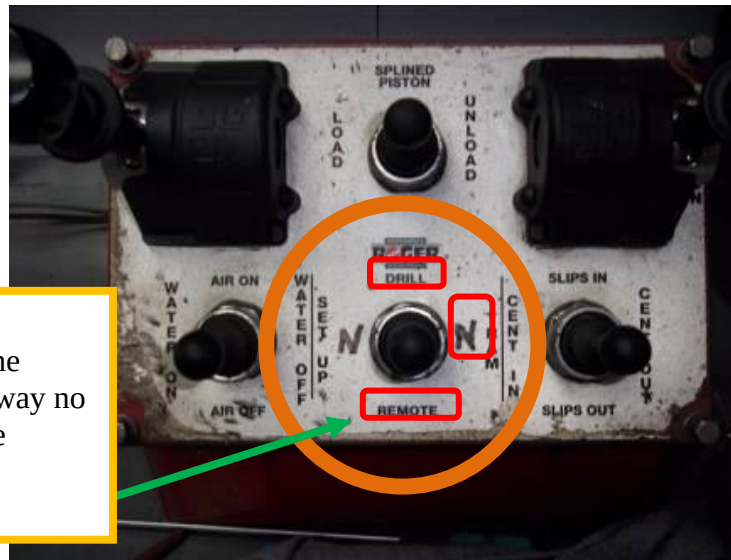


1. Slowly lower the gearbox to the wrench. If the wrench needs to be repositioned to align with the saver sub, it must be manipulated by the end so as not to expose itself to a jamming zone. (Yellow circle)
2. Once aligned, continue lowering the gearbox until it is firmly seated in the wrench.

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WARNING

Old console: When a worker must maintain, inspect or replace one or more components on the boom, it must necessarily position the joystick in central position (the neutral position).



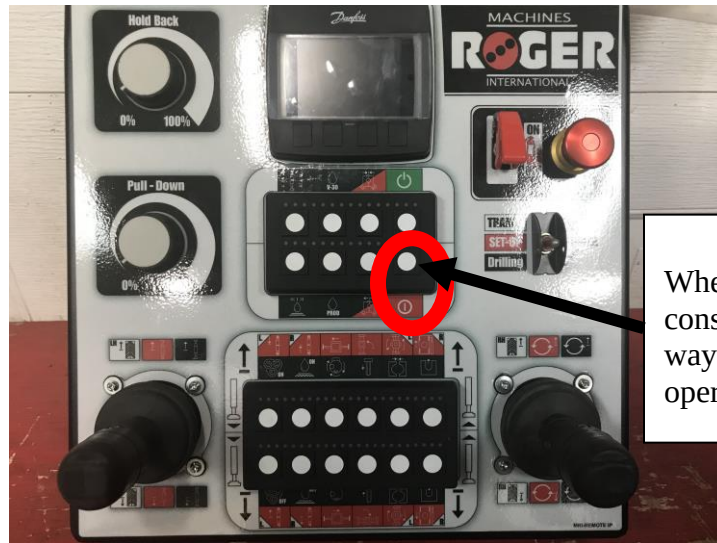
When this joystick is in the central position, power to the console is disabled. In this way no function of the boom can be operated on the drill.

5. Turn the rotation counterclockwise until the saver sub is unscrewed. Use the hydraulic force to completely unscrew the saver sub. It is possible that hits with the sledgehammer on the wrench will be needed;
6. Reassemble the gear box, remove the saver sub;
7. Remove the teflon protector from the saver sub to install on the new
8. Screw the new saver sub by hand for a few turns only.
9. Repeat steps 3 and 4 for approaching the gear box to the wrench.
10. Once the saver sub is inserted into the wrench, turn the rotation clockwise. Use hydraulic force, previously adjusted to 2800 psi to fully screw the sub saver to the contact gear box-saver sub.

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WARNING

New console: When a worker has to maintain, inspect or replace a component or components on the mast, he must necessarily press the stop button to remove power from the console



When this button is pressed, the console power is turned off. In this way no function of the boom can be operated on the drill

Adds or modifications	Date	By
Translation review	May 2024	S. Tremblay R, Harrower

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